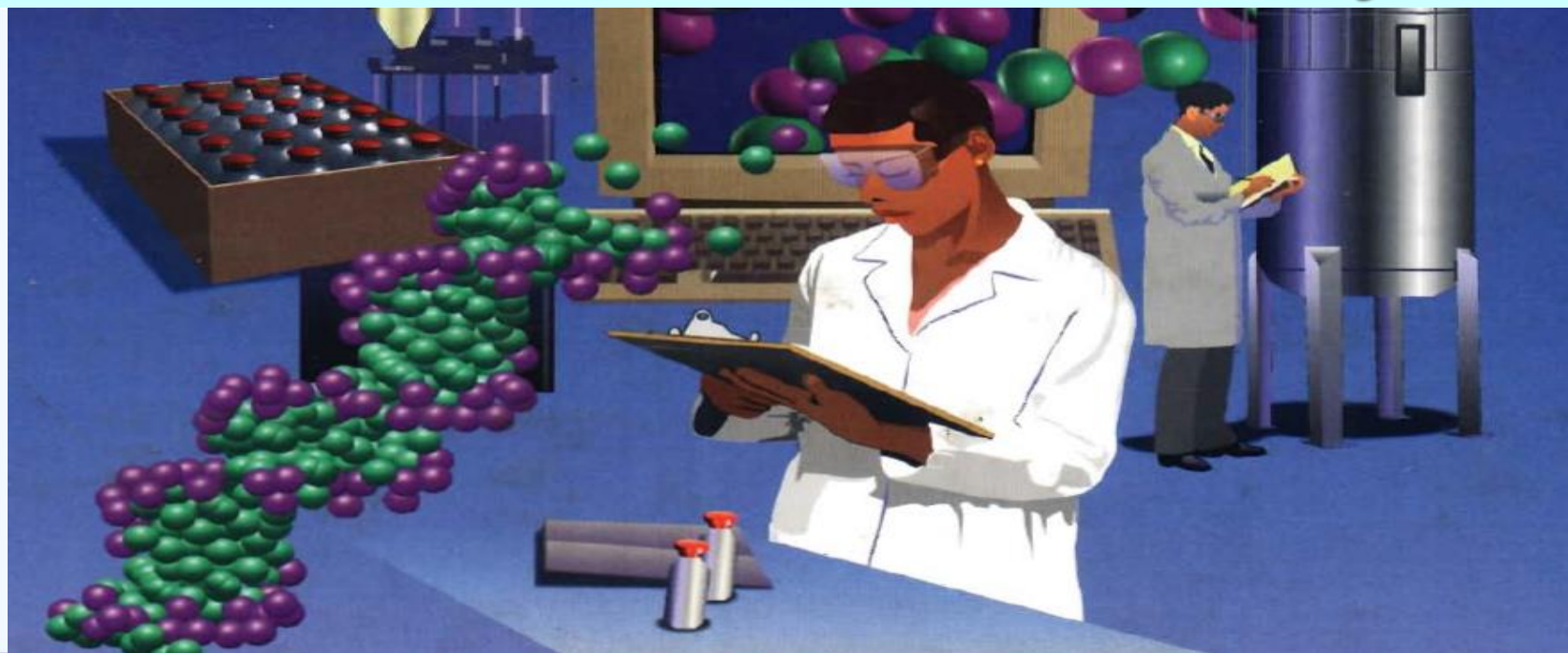


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现代分析测试技术 Modern Instrumental Analysis



课程特点

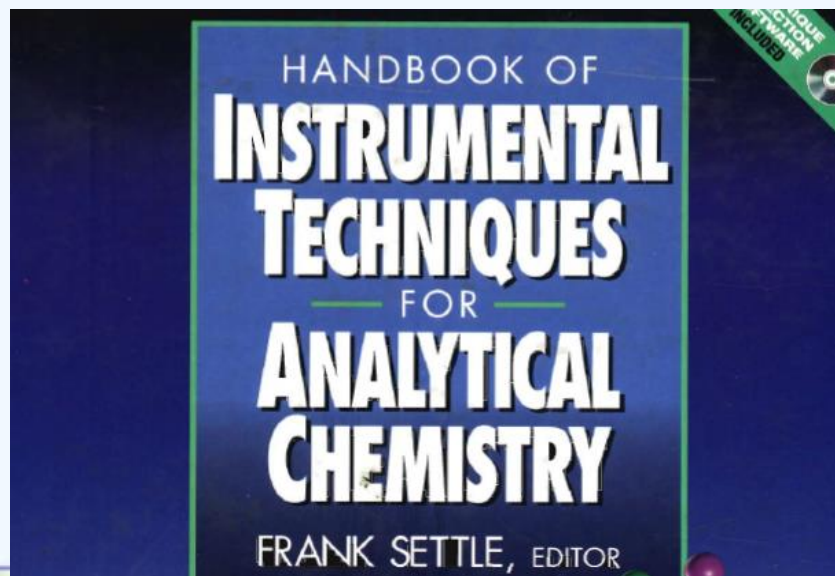
- 重要性：是海洋生物医药、化学、生化、环保、质检、食品安全等学科的必要工具
- 理论性：涉及GC、HPLC、CE GC/MS、LC/MS、IR、MS、NMR等
- 技术性：解决实际工作中各种分离分析和鉴定问题
- 前导课程：分析化学，有机化学

双语课程

推荐中文参考教材



推荐英文参考教材



- 1) 分析化学 Analytical Methodology
- 2) 样品前处理 Sample Preparation in Organic Analysis
- 3) 分离技术（色谱法） Separation Technology
- 4) 紫外-可见分光光度法 Ultraviolet-Visible Spectrophotometry
- 5) 红外光谱技术 Infrared Spectroscopic Analysis
- 6) 核磁共振波谱技术 Nuclear Magnetic Resonance Spectroscopy
- 7) 有机质谱技术 Organic Spectroscopy Technology
- 8) 大分子分析技术 Macromolecule Analysis
- 9) 其他分析方法 Other Analytical Methods

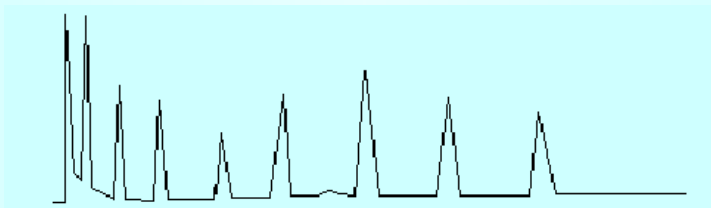
第一章

分析方法学

Analytical Methodology



理解和发现分析问题
Understanding and
Defining the Problem



下页



1) 分析问题的来由

History of the Sample and Background of the Problem

2) 文献查询Literature Search

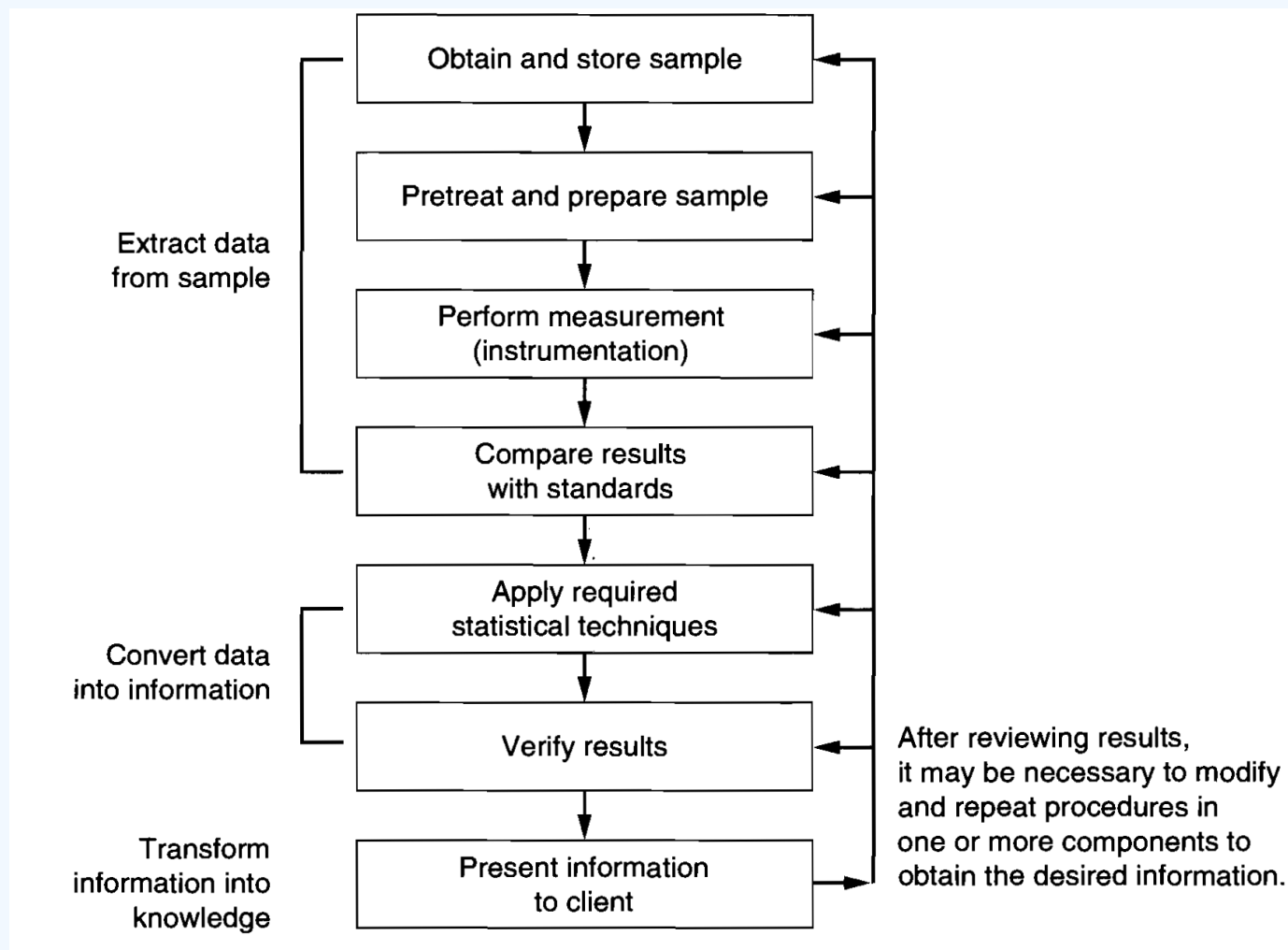
3) 制定分析计划Plan of Action and Execution

4) 未知成分的分析Approaches to Unknowns

5) 技术选择Technique Selection

分析方法的组成

Components of an analytical method

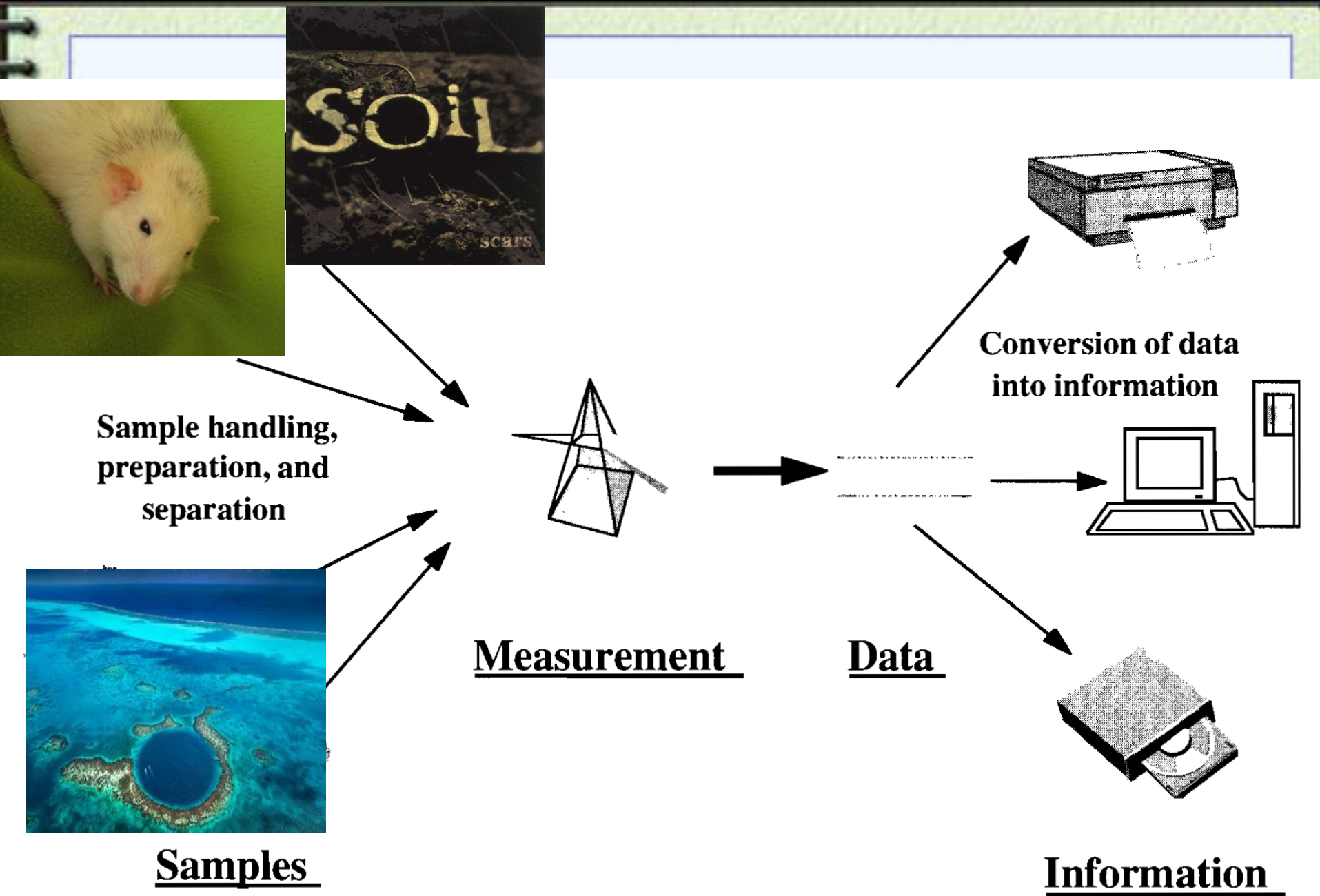


如何选择关键技术Key to technique selection (1)

Location of analyte	
B	Sample from bulk of material in question
S	Sample from or on surface of material in question
Physical state of sample	
G	Gas
L	Liquid (includes liquids dissolved in liquid solvents)
S	Solid (includes solids with gases adsorbed on surface)
Ds	Solid dissolved in appropriate liquid solvents
Dg	Gas dissolved in appropriate liquid solvent
Amount of sample	
L	Macro>1 milligram
S	Micro $\leq$ 1 milligram
Estimated sample purity	
P	Pure (>99%) element or compound
Sm	Simple mixture of up to six major components
M	Complex mixture

如何选择关键技术Key to technique selection (2)

Fate of sample	
D	Destructive analysis
N	Nondestructive analysis
Elemental information	
EI	Total analysis (element present in all chemical forms in sample)
Sp	Speciation (determination of individual elemental oxidation or chelation states)
Is	Isotopic and mass analysis
Molecular information	
Cp	Compounds present in sample
Io	Polyatomic ionic species present in sample
Fn	Functional group analysis
St	Structural analysis (structural and stereochemical determination)
Mw	Molecular weight determination
Pp	Physical property
Analysis type	
QI	Qualitative analysis
Qt	Quantitative analysis
Analyte concentration	
Mj	Major component determination (> 10 wt %)
Mn	Minor component determination (10 to 0.1 wt %)
2025/2/19Tr	Trace component determination (1 to 1000 ppm or 0.0001 to 0.1 wt %)
UI	Ultratrace component determination (<1 ppm or <0.0001 wt %)



Buyers'分析手册

- "LabGuide" published annually in August by Analytical Chemistry
- "Buyers' Guide" published annually in February by American Laboratory
- "Annual Directory" published in January by Laboratory Equipment
- Product reviews in the "A" pages of Analytical Chemistry focusing on a particular technique include general information on the selected technique and information on manufacturers

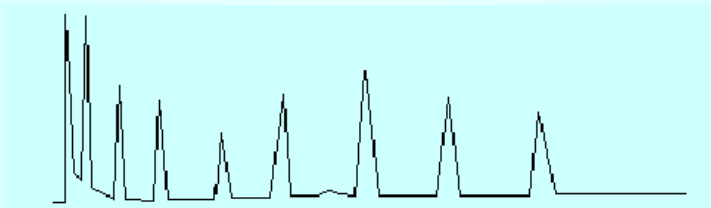
第二章

样品前处理Sample Preparation in Organic Analysis



所有分析的基础

Sampling Process



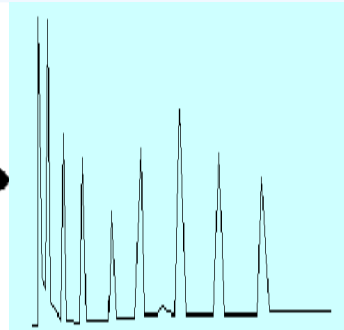
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分析常规步骤 Steps in the Analytical Cycle



Sample preparation



Sample analysis



Data handling



Sample
Transport and
storage



Archiving



Sample collection
and sampling

Information
to customer



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Report generation

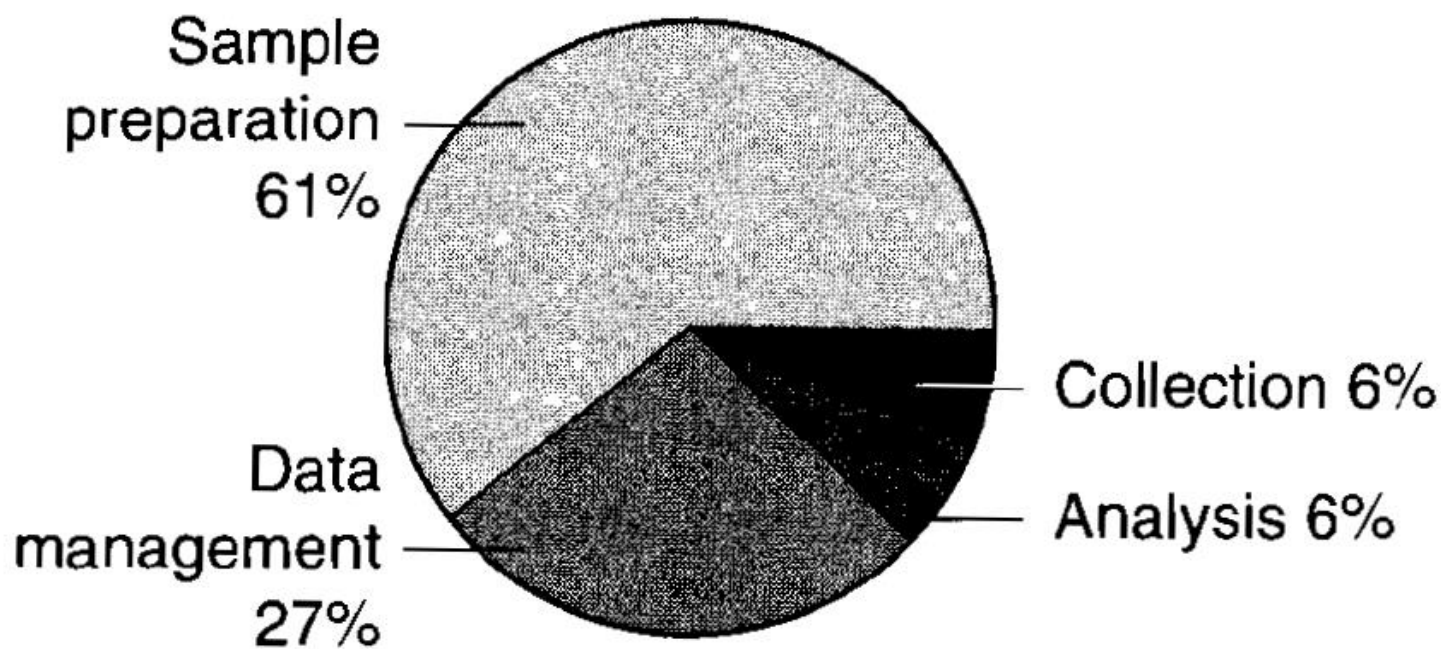
样品的转运和储存 Sample Transport and Storage

- 1) 选择合适的容器 Choice of appropriate sampling container
- 2) 加入抗氧化剂或杀菌剂 Addition of chemical stabilizers such as antioxidants and antibacterial agents
- 3) 冷冻样品 Freezing the sample to avoid thermal degradation
- 4) 样品吸附于支载材料 Adsorption on a solid phase

二次样品 Secondary Sampling

- 备份样品非常重要 Once the sample has made it to the laboratory, a representative subsample must be taken.
- 备份样品也叫二次样品 This process is called secondary sampling. Just as in primary sampling, the size or inhomogeneity of the sample may be a problem in secondary sampling.
- 规范二次样品流程可以减少分析数据的损失 statistically appropriate sampling procedures are applied to avoid discrimination, which can further degrade analytical data.

分析时间分布 Survey results for distribution of time analytical chemists spend on sample analysis.



(From R.E. Majors, LC/GC 9, pp. 16-20 (1991).)

样品前处理的目的

Goals and Objectives of Sample Preparation

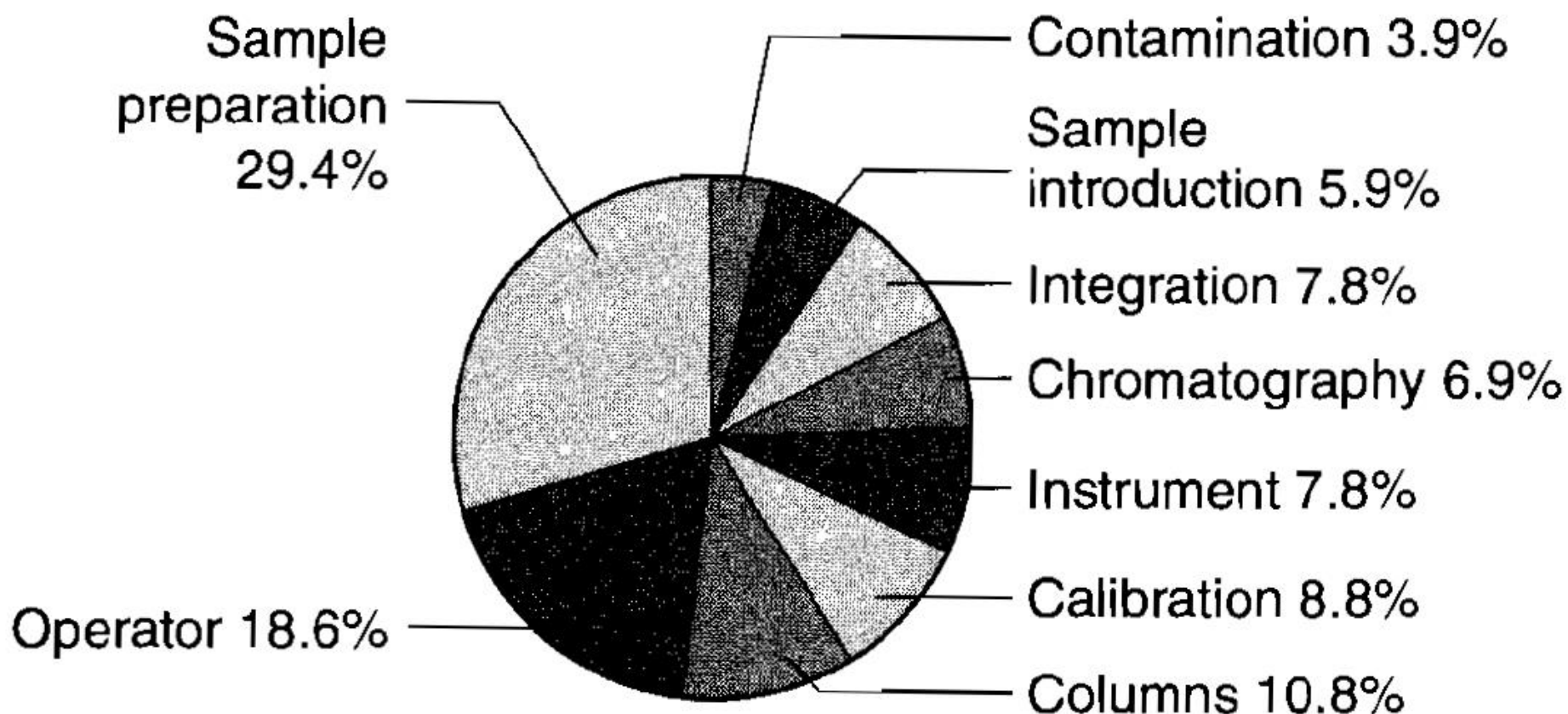
样品前处理的三个目标

Successful sample preparation has a threefold objective: namely, to provide the sample component of interest

- (1) 溶解 In solution
- (2) 防止干扰因素 Free from interfering matrix elements
- (3) 配成合适的浓度 At a concentration appropriate for detection and measurement

分析中误差来源比例

Survey results for the distribution of error generated during sample analysis.



(From R.E. Majors, LC/GC 9, pp. 16-20 (1991)).

样品前处理的初步分离和浓缩

Separation and Preconcentration Techniques in Sample Preparation

(1) 两项萃取法 The most common approach in analyte separation involves a two-phase system where the analyte and interferences are distributed between the two phases.

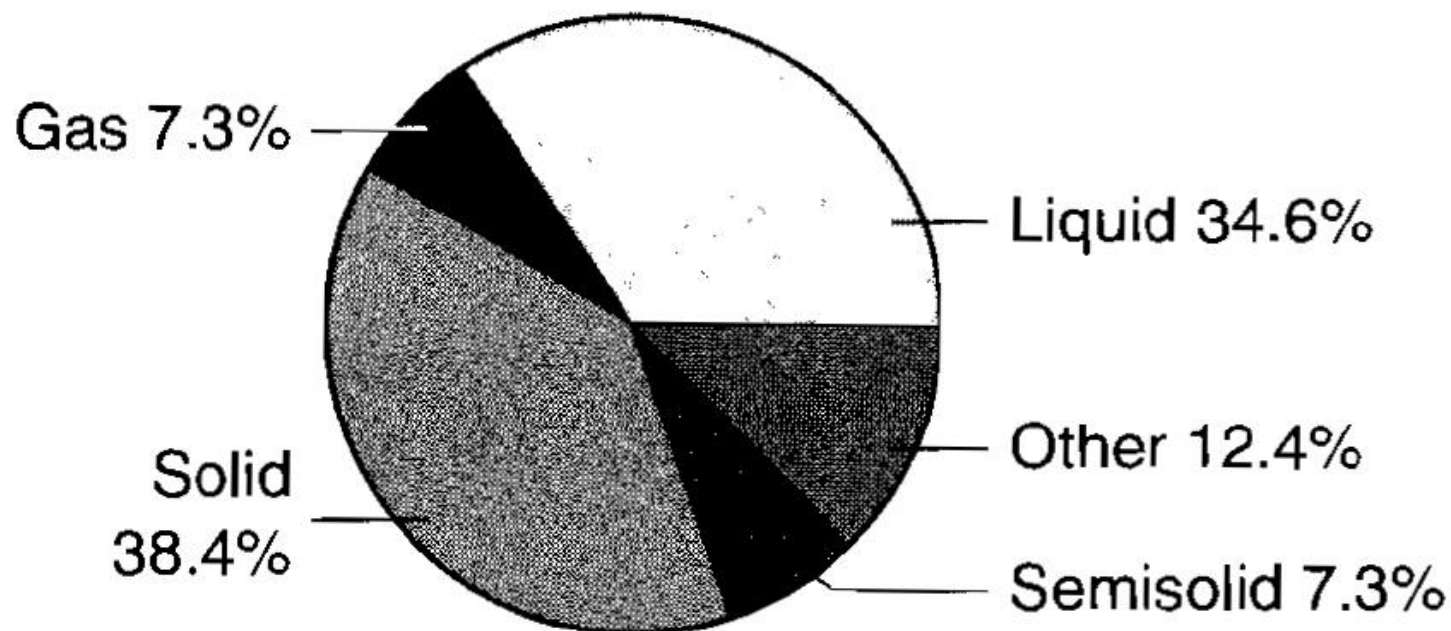
Distribution is an equilibrium

(2) 过滤和蒸馏法 Filtration and Distillation

(3) 固相萃取法 that the analyte has a very high retention. A large volume of sample is passed through the column to concentrate the analyte on the solid phase. Finally, a strong solvent is used to remove the concentrated analyte.

分析样品来源分布

Distribution of sample types



((From R.E. Majors, LC/GC 9, pp. 16-20 (1991).

传统固体样品的提取方法

Traditional Extraction Methods for Solid Samples

前处理方法 Method of Sample Pretreatment	Principles of Technique
固液提取 Solid-liquid extraction	Sample is placed in stoppered container and solvent is added that dissolves analyte of interest; solution is separated from solid by filtration (sometimes called shake/filter method).
索氏提取 Soxhlet extraction	Sample is placed in disposable porous container (thimble); constantly refluxing solvent flows through the thimble and leaches out analytes, which are continuously collected in boiling pot.
均质化 Homogenization	Sample is placed in a blender, solvent is added, and sample is homogenized to a finely divided state; solvent is removed for further workup.
声波降解法 Sonication	Finely divided sample is immersed in ultrasonic bath with solvent and subjected to ultrasonic radiation. An ultrasonic probe or ultrasonic cell disrupter can also be used.
溶解 Dissolution	Sample is treated with solvent and taken directly into solution with or without chemical change.

现代固体样品的提取方法

Modern Extraction Methods for Solid Samples

Method of Sample Pretreatment	Principles of Technique
加强型溶剂提取 Accelerated (enhanced) solvent extraction (ASE or ESE)	Sample is placed in a sealed container and heated to above its boiling point, causing pressure in vessel to rise; extracted sample is automatically removed and transferred to vial for further treatment.
自动索氏提取 Automated Soxhlet extraction	A combination of hot solvent leaching and Soxhlet extraction; sample in thimble is first immersed in boiling solvent, then thimble is raised for Soxhlet extraction with solvent refluxing.
超临界流体提取 Supercritical fluid extraction (SFE)	Sample is placed in flow-through container and supercritical fluid (such as CO ₂) is passed through sample; after depressurization, extracted analyte is collected in solvent or trapped on adsorbent, followed by desorption by rinsing with solvent.
微波辅助提取 Microwave-assisted extraction (MASE)	Sample is placed in an open or closed container and heated by microwave energy, causing extraction of analyte.

典型固体样品处理方法的比较

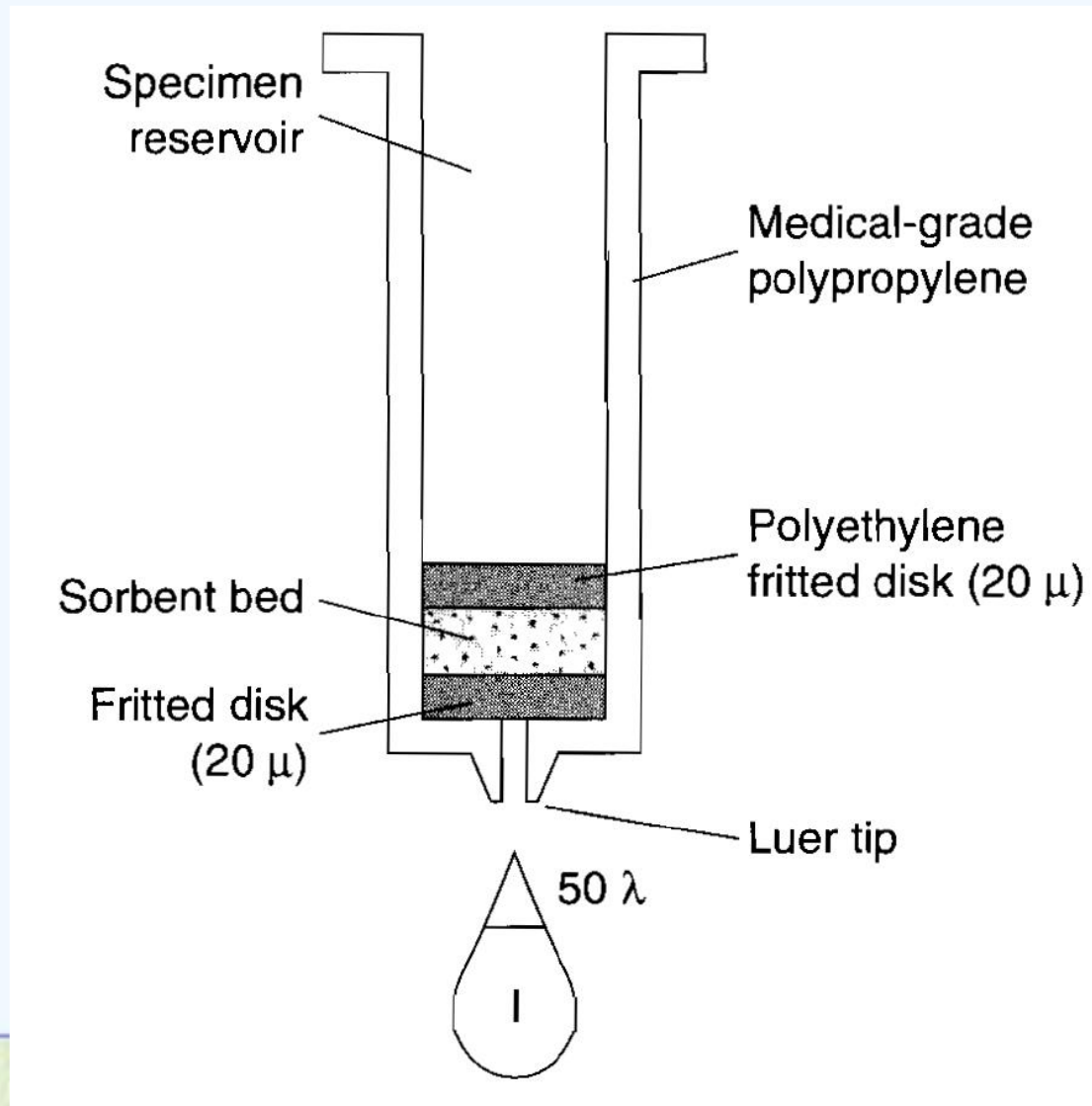
Comparison of Extraction Methods for Sample Preparation of Solids

Parameter	Sonication	Soxhlet (traditional)	Soxhlet (modern)	SFE	ASE (ESE)	Microwave- Assisted (closed container)	Microwave- Assisted (open container)
Sample size, g	20-50	10-20	10-20	5-10	5-15	2-5	2-10
Solvent volume, ml	100-300	200-500	50-100	10-20	10-15	30	20-30
Temperature, °C	Ambient-40	40-100	40-100	50-150	50-200	100-200	Ambient
Pressure	Atmospheric	Atmospheric	Atmospheric	2000- 4000psi	1500- 2000psi	1500-2000psi	Atmospheric
Time, hr	0.5-1.0	12-24	1-4	0.5-1.0	0.2-0.3	0.2-0.3	0.1-0.2
Degree of automation	0	0	++	+++	+++	++	++
Number of samples	High	1	6	44	24	12	12
Cost	Low	Very low	Moderate	High	High	Moderate	Moderate

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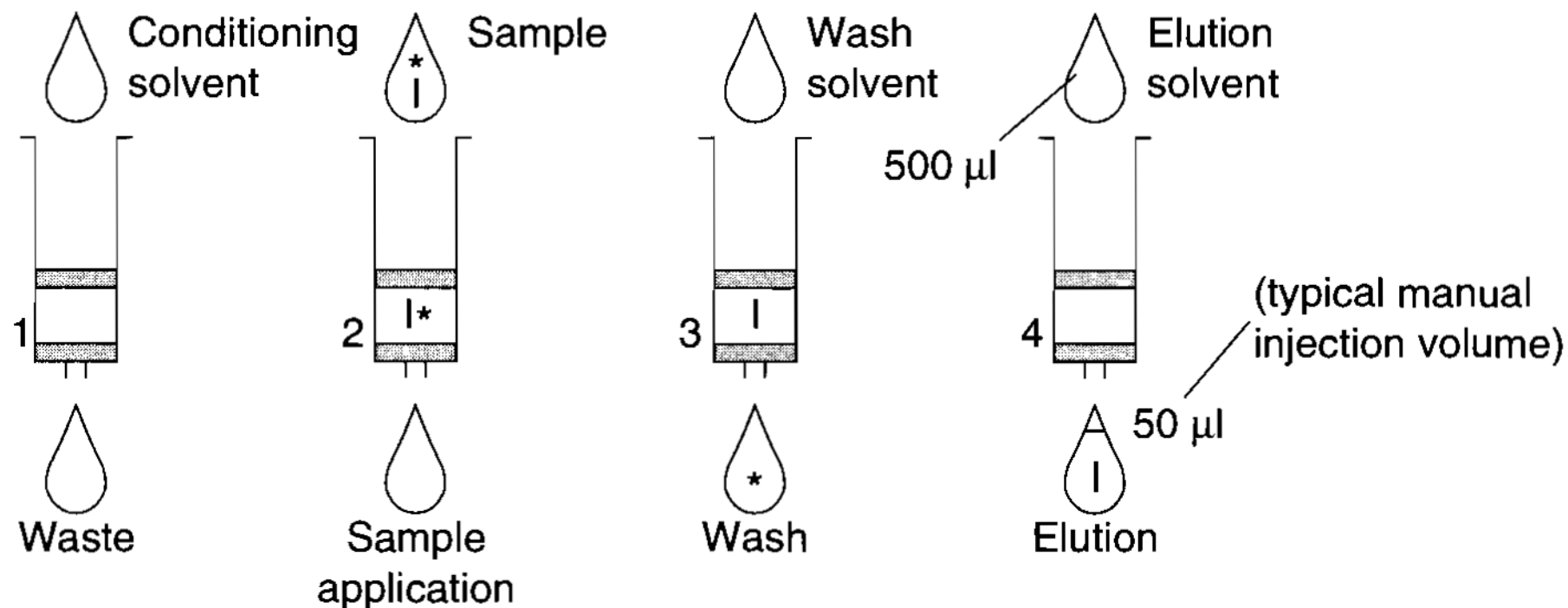
液体样品前处理

Sample Preparation Methods for Liquid Samples



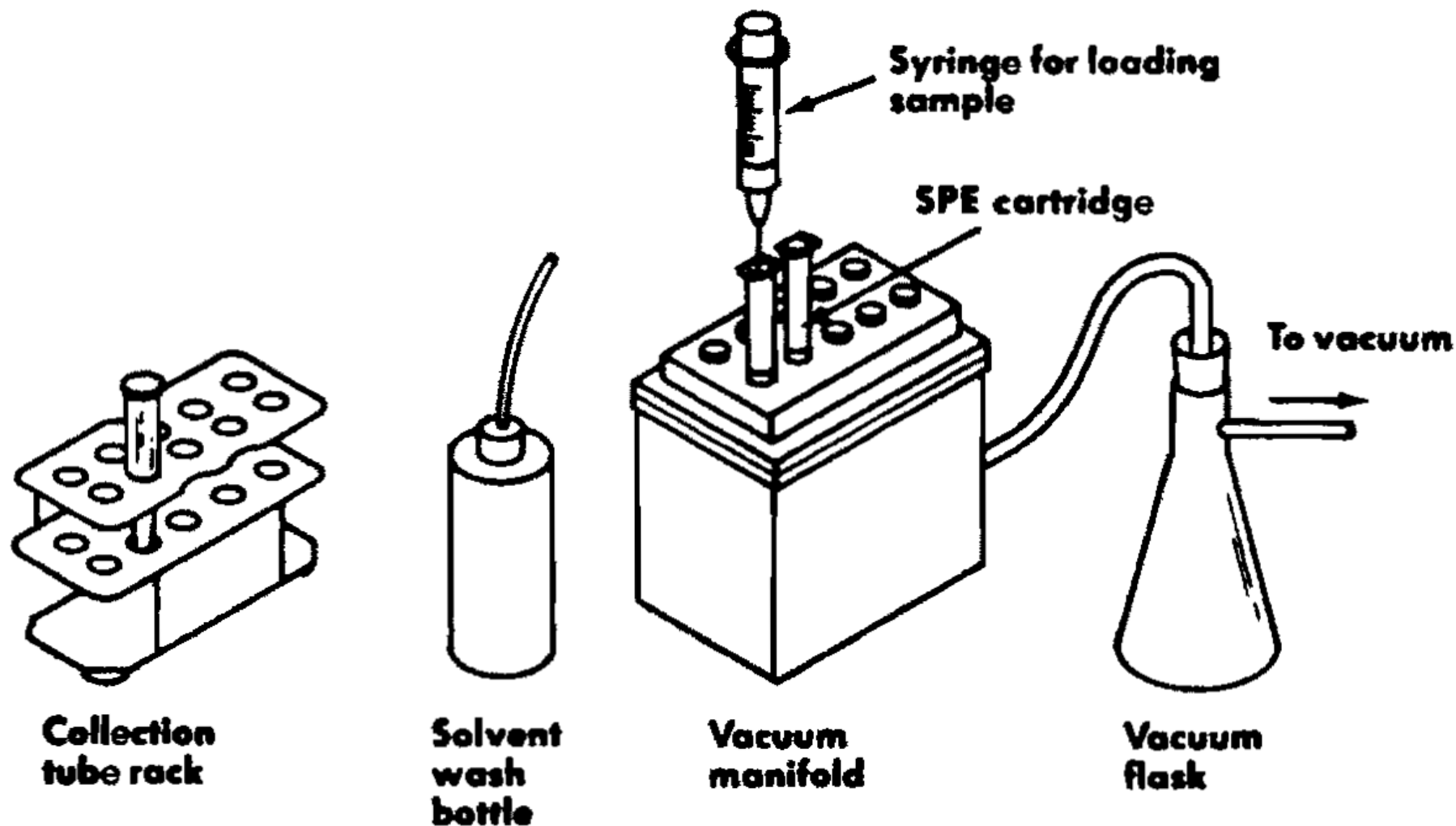
固相萃取步骤

Steps in the Operation of s Solid-Phase Extraction Experiment



固相萃取设备 Apparatus for Solid-Phase Extraction

Steps in the Operation of s Solid-Phase Extraction Experiment



思考题

- 海洋有巨大的稀释作用，如何富集近海有机污染物？
- Why is the sample pretreatment so important?